

DHRUV DAVE

Toronto, ON

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Education

Queen's University

Bachelor of Computer Science (Honours), Mathematics Minor | GPA: 3.8

September 2024 – Present

Kingston, ON

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, SQL

Frameworks & Libraries: React, FastAPI, Node.js, Hono, Flask, SQLAlchemy, Tailwind CSS, Pandas, NumPy

Databases: PostgreSQL, Redis, SQLite

Developer Tools: Git, Docker, Bash, Power BI, Chrome Extension APIs

Experience

Cerebro AI Technical Specialist | FGF Brands

May 2026 – August 2026

Data & Platform Engineering

- Built internal tooling in Node.js and Python documenting Cerebro's OEE telemetry models, changeover logic, and troubleshooting decision trees; developed Power BI dashboards fed by Cerebro's production data to track manufacturing performance across FGF's 22 facilities.
- Write and optimize SQL diagnostic queries against Cerebro's production database to detect and reconcile data anomalies; work to date has identified systemic misconfigurations in workflow state transitions and upstream pipeline gaps across over 2,000 production records.
- Won 1st Place at FGF's 4th Annual FGFLife Hackathon**, building BakeryInsight (React 19, TypeScript, Node.js, Express), a full-stack trend-intelligence dashboard that classifies emerging food trends across 9 data sources (Google Trends, TikTok, Reddit, Instagram, Mintel, and others), scores them on growth, saturation, and regional spread, and pairs each with a 6-dimension feasibility engine (manufacturing, supply chain, cost, safety, brand, shelf life) scored against real FGF production-line and ingredient data.

Research Assistant | COBWEB @ University of Toronto

Jan 2024 – August 2024

Computational Simulation & Data Analysis

- Implemented agent-based simulations on the COBWEB platform (Dr. Brad Bass, UofT) to model emergent behaviour in ecological adaptive systems; configured multi-condition experiments comparing control and experimental agent populations, with results used in the team's research.
- Wrote Python data-processing pipelines using Pandas and NumPy to clean, normalize, and visualize multi-run simulation outputs; the analysis revealed population dynamics and threshold-crossing patterns across experimental conditions.

Software Engineer Intern | Digitera

Sep 2022 – Feb 2023

Web Application Development

- Developed and shipped web applications for small business clients using JavaScript, HTML, and CSS; optimized page performance by 40% through asset minification, image compression, and deferred script loading.

Projects

Markr — Stock Screener | React 19, FastAPI, PostgreSQL, Redis, Docker | [github](#) | [live](#)

- Built a full-stack stock screener with a FastAPI backend, Redis cache, and PostgreSQL for range-partitioned price history; implemented a 6-source data fallback chain (Finnhub → Yahoo Finance → FMP → Alpha Vantage → Stooq → Twelve Data) with per-source rate limiting enforced via an atomic Redis Lua script, across 12 REST endpoints.
- Computed RSI(14), MACD(12/26/9), Bollinger Bands(20), MA50, and MA200 client-side from raw price history; candle data writes asynchronously to a monthly-partitioned Postgres table after each fetch, with Redis TTLs configurable per resolution (60s intraday, 3600s daily).
- Wrote a Vitest suite covering the fallback chain logic, indicator calculations, and rate limiter behavior; portfolio positions and price alerts persist in the browser via the Web Notifications API — frontend deployed on Vercel with GitHub Actions CI.

Password Security Analyzer & Manager | React 18, Flask, Python, SQLite, Docker | [github](#) | [live](#)

- Built a password security analyzer that computes entropy, detects keyboard walks, date sequences, and common password patterns, and scores strength 0–100 with improvement recommendations; vault entries are encrypted with AES-256-GCM using per-entry keys derived via PBKDF2 at 600,000 iterations.
- Secured user accounts with bcrypt (cost factor 12), JWT access/refresh token rotation, TOTP-based 2FA (pyotp with QR code provisioning), and rate limiting; integrated HaveIBeenPwned breach checking via SHA-1 prefix k-anonymity so full hashes are never transmitted.
- Implemented 18 REST endpoints across auth, vault, password tools, and a security dashboard; containerized the stack in separate Docker images behind nginx, and wrote a pytest suite covering auth, encryption, and breach-check modules.

ctx — Browser Session Tracker | TypeScript, Node.js, Hono, SQLite, Chrome MV3 | [github](#)

- Built a local-first browser session tracker as a TypeScript npm workspace monorepo across three packages — Chrome MV3 extension, Hono/Node.js backend, and a CLI; sessions batch via `chrome.alarms` and persist to `chrome.storage.local` without a running server.
- Wrote a privacy guard in the content script that checks URL patterns, a hardcoded sensitive-domain list, and live DOM fields (password inputs, credit card autocomplete) before capturing anything; blocked pages are logged with no URL, title, or content.